

Class 08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from final needle aspiration (FNA) of breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`.

```
# Input data file is in Project directory
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)

head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
84348301	0.14250	0.28390	0.2414		0.10520
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419		0.07871	1.0950	0.9053
842517	0.1812		0.05667	0.5435	0.7339
84300903	0.2069		0.05999	0.7456	0.7869
84348301	0.2597		0.09744	0.4956	1.1560
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003		0.006193	25.38	17.33
842517	0.01389		0.003532	24.99	23.41
84300903	0.02250		0.004571	23.57	25.53
84348301	0.05963		0.009208	14.91	26.50
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0		0.1622	0.6656
842517	158.80	1956.0		0.1238	0.1866
84300903	152.50	1709.0		0.1444	0.4245
84348301	98.87	567.7		0.2098	0.8663
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119		0.2654	0.4601	
842517	0.2416		0.1860	0.2750	
84300903	0.4504		0.2430	0.3613	
84348301	0.6869		0.2575	0.6638	
	fractal_dimension_worst				
842302		0.11890			

842517	0.08902
84300903	0.08758
84348301	0.17300

Make sure we remove or exclude the **diagnosis** column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
# Use -1 to omit the first (diagnosis) column.
wisc.data <- wisc.df[,-1]

# Store first column, diagnosis, as a factor.
diagnosis <- as.factor(wisc.df$diagnosis)

head(wisc.data, 4)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
842302	17.99	10.38	122.80	1001.0	0.11840
842517	20.57	17.77	132.90	1326.0	0.08474
84300903	19.69	21.25	130.00	1203.0	0.10960
84348301	11.42	20.38	77.58	386.1	0.14250

	compactness_mean	concavity_mean	concave.points_mean	symmetry_mean
842302	0.27760	0.3001	0.14710	0.2419
842517	0.07864	0.0869	0.07017	0.1812
84300903	0.15990	0.1974	0.12790	0.2069
84348301	0.28390	0.2414	0.10520	0.2597

	fractal_dimension_mean	radius_se	texture_se	perimeter_se	area_se
842302	0.07871	1.0950	0.9053	8.589	153.40
842517	0.05667	0.5435	0.7339	3.398	74.08
84300903	0.05999	0.7456	0.7869	4.585	94.03
84348301	0.09744	0.4956	1.1560	3.445	27.23

	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	0.006399	0.04904	0.05373	0.01587
842517	0.005225	0.01308	0.01860	0.01340
84300903	0.006150	0.04006	0.03832	0.02058
84348301	0.009110	0.07458	0.05661	0.01867

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656

842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

There are 569 observations (patient samples) in the dataset.

Q2. How many of the observations have a malignant diagnosis?

```
# Access the diagnosis column of the original data frame.
# Use table() function to create frequency table
table(wisc.df$diagnosis)
```

```

  B    M
357 212

```

```
# We can also use the sum() function and a logical argument.
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

212 observations have a malignant diagnosis.

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function (performs pattern matching in a character vector) to help us here:

```
length(grep("_mean", colnames(wisc.data), value = T))
```

```
[1] 10
```

10 variables/features in the data are suffixed with `_mean`.

Principle Component Analysis (PCA)

Performing PCA

We first need to check if the data needs to be scaled prior to performing PCA. Scaling is especially necessary if the means (reflecting units of measurement) and/or standard deviations (reflecting variance) of the columns/features are very different.

```
# Check means of wisc.data columns.  
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
# Check standard deviations of wisc.data columns.
apply(wisc.data, 2, sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

We should scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010

Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

44.27% of the original variance is captured by PC1.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 PCs are required to describe at least 70% of the original variance (at PC3, the cumulative proportion is 72.636%).

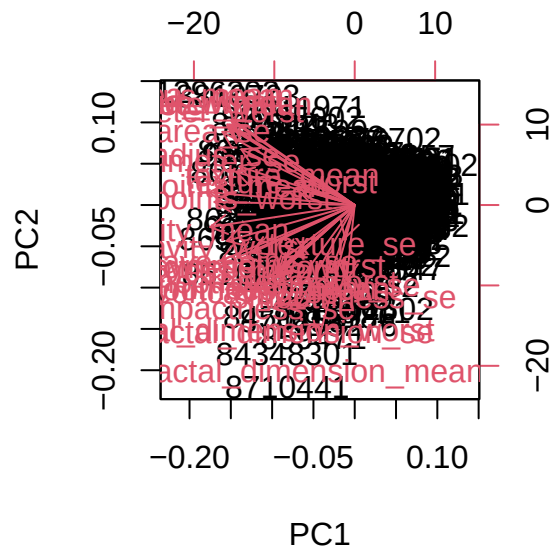
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 PCs are required to describe at least 90% of the original variance (at PC7, the cumulative proportion is 91.010%).

Interpreting PCA results

Create a biplot of the `wisc.pr` using the `biplot()` function.

```
biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

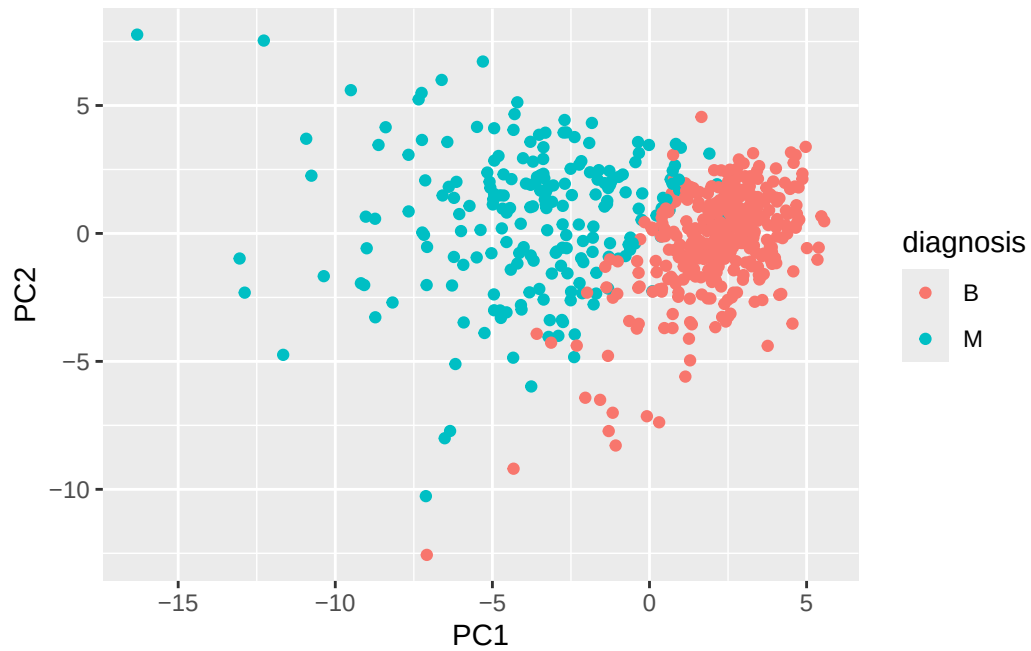
A few things stand out to me from the biplot. Firstly, the observations/biopsy samples are plotted by their row names (in this case, sample ids). Secondly, in addition to the observations themselves in black, the plot also includes red vectors for each of the variables (each vector shows the contribution of a given variable/biopsy feature to PC1 and PC2). Altogether, the plot is very dense.

The plot is difficult to understand, both due to using rownames as the plotting character (individual observations cannot be distinguished) and due to the overlay of so many labelled, variable vectors. We cannot identify sample groupings/clusters or patterns.

Let's see the **PC score plot** of PC1 vs PC2, and color the points by the diagnosis.

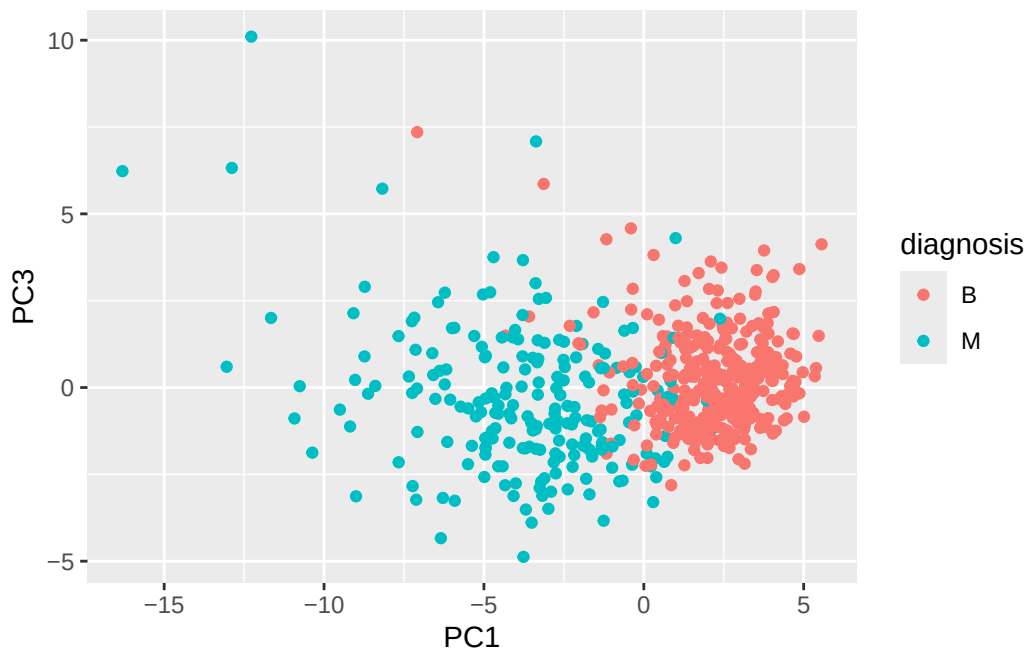
```
library(ggplot2)

# Scatter plot of observations along principle components 1 and 2.
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
# Repeat for components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col = diagnosis) +
  geom_point()
```



The PC score plots clearly show separation between the benign (red) and malignant (turquoise) observations/biopsy samples along the *PC1 axis* (more so than along the PC2 or PC3 axes). Benign samples have more positive PC1 scores than malignant samples.

Variance explained

Calculate the variance of each principle component.

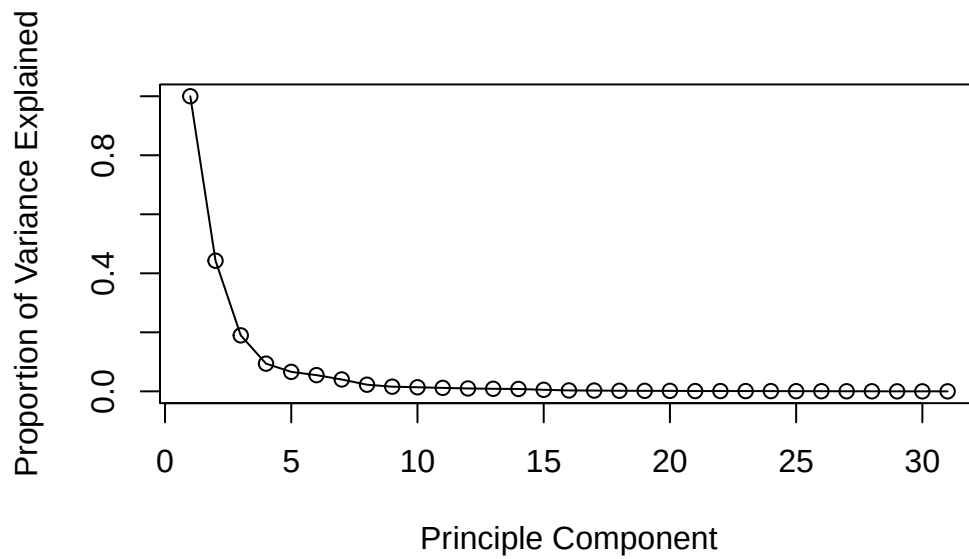
```
# Calculate the variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Calculate the variance explained by each principle component.

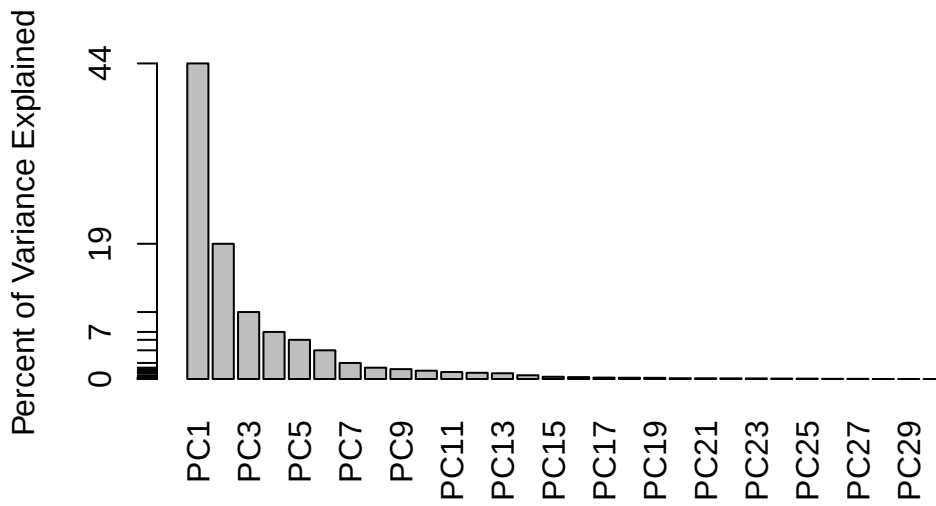
```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for by each PC.
plot(c(1, pve), xlab = "Principle Component", ylab = "Proportion of Variance Explained", ylim = c(0, 1))
```



We can also depict the scree plot as a bar plot.

```
# Alternative scree plot (bar plot) of the same data, of variance explained by each PC.  
barplot(pve, ylab = "Percent of Variance Explained", names.arg = paste0("PC", 1:length(pve)))  
axis(2, at=pve, labels=round(pve,2)*100)
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
# Find PC1 loading value for concave.points_mean
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

```
# Following returns TRUE for features with larger contributions to PC1 than concave.points_mean
abs(wisc.pr$rotation[,1]) > abs(wisc.pr$rotation["concave.points_mean",1])
```

radius_mean	texture_mean	perimeter_mean
FALSE	FALSE	FALSE
area_mean	smoothness_mean	compactness_mean
FALSE	FALSE	FALSE
concavity_mean	concave.points_mean	symmetry_mean
FALSE	FALSE	FALSE

fractal_dimension_mean	radius_se	texture_se
FALSE	FALSE	FALSE
perimeter_se	area_se	smoothness_se
FALSE	FALSE	FALSE
compactness_se	concavity_se	concave.points_se
FALSE	FALSE	FALSE
symmetry_se	fractal_dimension_se	radius_worst
FALSE	FALSE	FALSE
texture_worst	perimeter_worst	area_worst
FALSE	FALSE	FALSE
smoothness_worst	compactness_worst	concavity_worst
FALSE	FALSE	FALSE
concave.points_worst	symmetry_worst	fractal_dimension_worst
FALSE	FALSE	FALSE

```
# sum() counts number of Ts; if 0, no features contribute more to PC1 than concave.points_mean
sum(abs(wisc.pr$rotation[,1]) > abs(wisc.pr$rotation["concave.points_mean",1]))
```

```
[1] 0
```

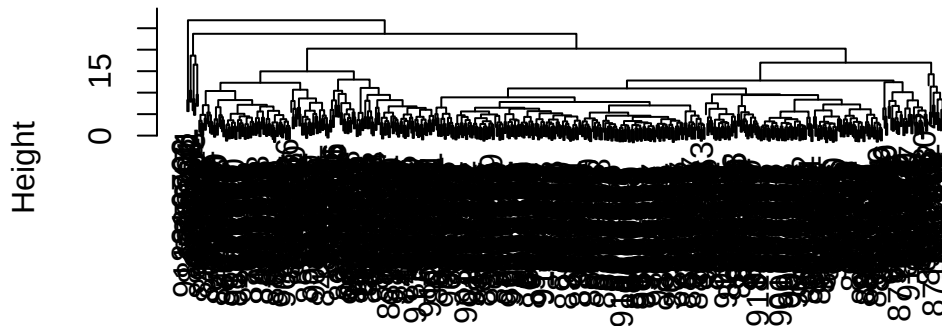
For PC1, the component of the loading vector for `concave.points_mean` is -0.2608538. There are no features with larger contributions to PC1 than this component.

Hierarchical Clustering

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
# Calculate Euclidean distances between all pairs of scaled observations
data.dist <- dist(data.scaled)
# Create a hierarchical clustering model using complete linkage.
wisc.hclust <- hclust(data.dist, method = "complete")

plot(wisc.hclust)
```

Cluster Dendrogram



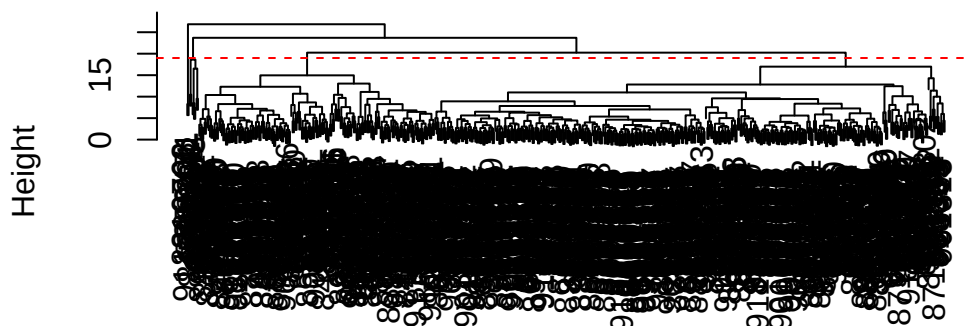
```
data.dist  
hclust (*, "complete")
```

Results of hierarchical clustering

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)  
abline(h = 19, col = "red", lty = 2)
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

The clustering model has 4 clusters at about $h = 19$.

Selecting number of clusters

```
# Cut tree to return 4 clusters.
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1  2  3  4
177  7 383  2
```

```
# Use table() to compare cluster membership and diagnoses.
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B   M
1      12 165
2       2   5
3     343  40
4       0   2
```

Cluster 1 mostly contains malignant observations, and Cluster 3 mostly contains benign observations.

Using different methods

Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

```
# Generate hierarchical clustering models using single, average, and ward.D2 linkage methods
wisc.hclust.single <- hclust(data.dist, method = "single")
wisc.hclust.single.clusters <- cutree(wisc.hclust.single, k = 4)

wisc.hclust.avg <- hclust(data.dist, method = "average")
wisc.hclust.avg.clusters <- cutree(wisc.hclust.avg, k = 4)

wisc.hclust.ward <- hclust(data.dist, method = "ward.D2")
wisc.hclust.ward.clusters <- cutree(wisc.hclust.ward, k = 4)

# Determine which method returns my favorite results.
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis		
wisc.hclust.clusters	B	M	
1	12	165	
2	2	5	
3	343	40	
4	0	2	

```
table(wisc.hclust.single.clusters, diagnosis)
```

	diagnosis		
wisc.hclust.single.clusters	B	M	
1	356	209	
2	1	0	
3	0	2	
4	0	1	

```
table(wisc.hclust.avg.clusters, diagnosis)
```


	diagnosis	
wisc.hclust.avg.clusters	B	M
1	355	209
2	2	0
3	0	1
4	0	2

```
table(wisc.hclust.ward.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.ward.clusters	B	M
1	0	115
2	6	48
3	337	48
4	14	1

The “complete” method generates my favorite results. I ran hierarchical clustering for the scaled data using complete, single, average, and ward.D2 linkage methods, and cut all trees into $k=4$ clusters. To evaluate the quality of the results, I compared cluster membership to actual diagnoses for each method; for me, the ideal result would have one cluster containing all benign samples and one cluster containing all malignant samples.

The “single” and “average” methods were my least favorite, each resulting in one cluster containing almost all samples (both malignant and benign).

The “ward.D2” method performed better, with cluster 1 having 115 of the 212 malignant samples (and 0 false positives) and cluster 3 having 337 of 357 benign samples (though 48 false negatives). However, the “complete” method seemed to perform best in terms of (1) maximizing the number of benign samples captured in one cluster and malignant samples in another, and (2) minimizing the number of false negatives (which I consider a much bigger and riskier flaw than false positives). The “complete” method’s cluster 1 captured 165 of the 212 malignant samples (compared to 115 in ward.D2) and cluster 3 captured 343 of the 357 benign samples (compared to 337 in ward.D2, and with fewer—40—false negatives).

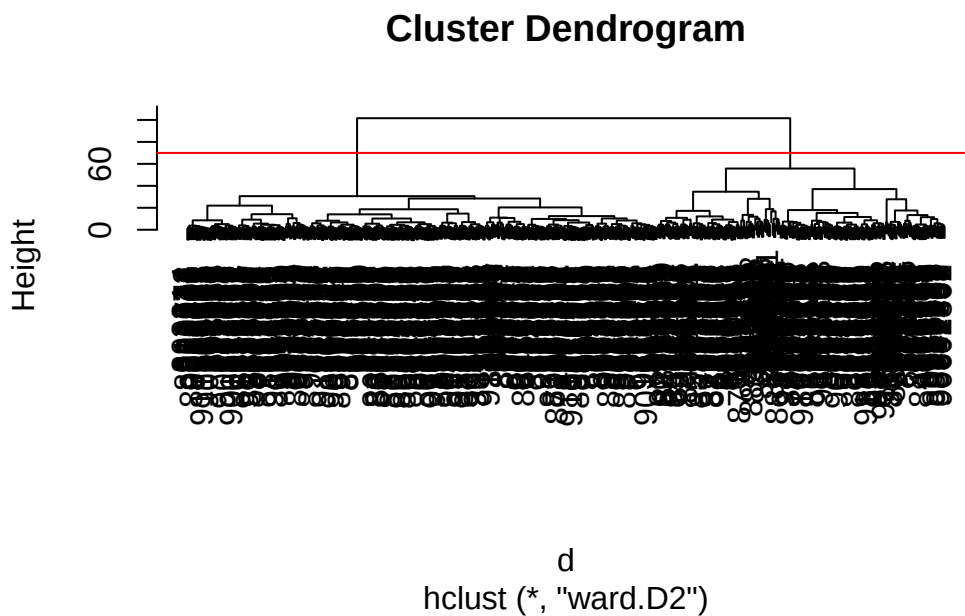
Combining methods

Clustering on PCA results

Create a hierarchical clustering model with linkage method=“ward.D2”, using the minimum number of principal components required to describe at least 90% of the variability in the data (previously determined to be PC1 to PC7).

```
d <- dist(wisc.pr$x[, 1:7])
wisc.pr.hclust <- hclust(d, method = "ward.D2")

plot(wisc.pr.hclust)
abline(h=70, col="red")
```



```
# Cut dendrogram into its two main branches.
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
table(wisc.pr.hclust.clusters)
```

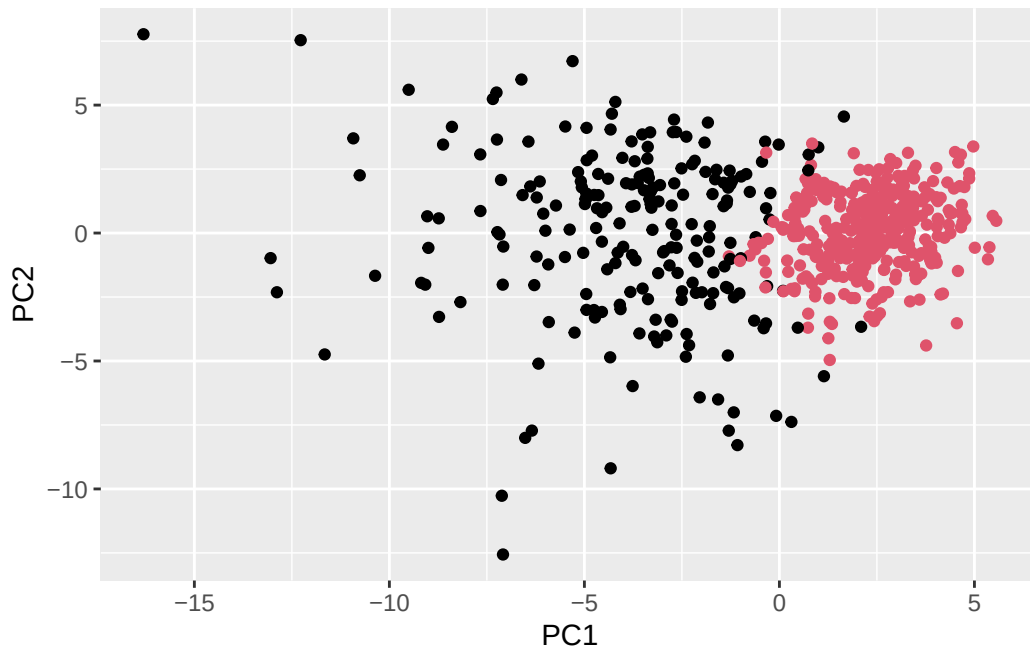
```
wisc.pr.hclust.clusters
  1  2
216 353
```

How do this clustering's `wisc.pr.hclust.clusters` correspond to the expert diagnosis?

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```
          diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24
```

```
# Color PC score plot of PC1 vs. PC2 by hclust groups.
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col = wisc.pr.hclust.clusters)
```



Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

It appears that cluster 1 is the malignant cluster (with 28 false positives, 188 true positives) and cluster 2 is the benign cluster (329 true negatives, 24 false negatives). By creating distinct “M” and “B” clusters, the PCA-based hclust model does fairly well at separating out the two diagnoses. However, we might want to focus on reducing the number of false negatives in the second, benign cluster, since these are malignant samples observed incorrectly clustered as benign.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
# Outputs of PCA-based and PCA-independent models, respectively.
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B   M
1    28 188
2   329  24

```

```
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B   M
1    12 165
2     2   5
3   343  40
4     0   2

```

The hierarchical clustering model generated without PCA *did* generate a distinct malignant cluster (165 positives, 12 false positives) and benign cluster (343 negatives, 40 false negatives). However, importantly, it yields more false negatives (40) than the model generated with PCA (24).

It is important to minimize the number of false negatives, since these are malignant samples incorrectly clustered as benign. Since the PCA-dependent model (`wisc.pr.hclust.clusters`) produced fewer false negatives than the PCA-independent model (`wisc.hclust.clusters`), I believe the latter is comparatively less successful at separating the diagnoses.

Prediction

Use the `predict()` function to apply PCA model to a new cancer cell dataset.

```

#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)

# Apply PCA model to new dataset, and print results.
npc <- predict(wisc.pr, newdata=new)
npc

```

```

      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10      PC11      PC12      PC13      PC14
[1,] -0.2307350  0.1029569 -0.9272861  0.3411457  0.375921  0.1610764  1.187882

```

```

[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
      PC15      PC16      PC17      PC18      PC19      PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153 0.1448061 -0.40509706 0.06565549 0.25591230 -0.4289500
      PC21      PC22      PC23      PC24      PC25      PC26
[1,] 0.1228233 0.09358453 0.08347651 0.1223396 0.02124121 0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
      PC27      PC28      PC29      PC30
[1,] 0.220199544 -0.02946023 -0.015620933 0.005269029
[2,] -0.001134152 0.09638361 0.002795349 -0.019015820

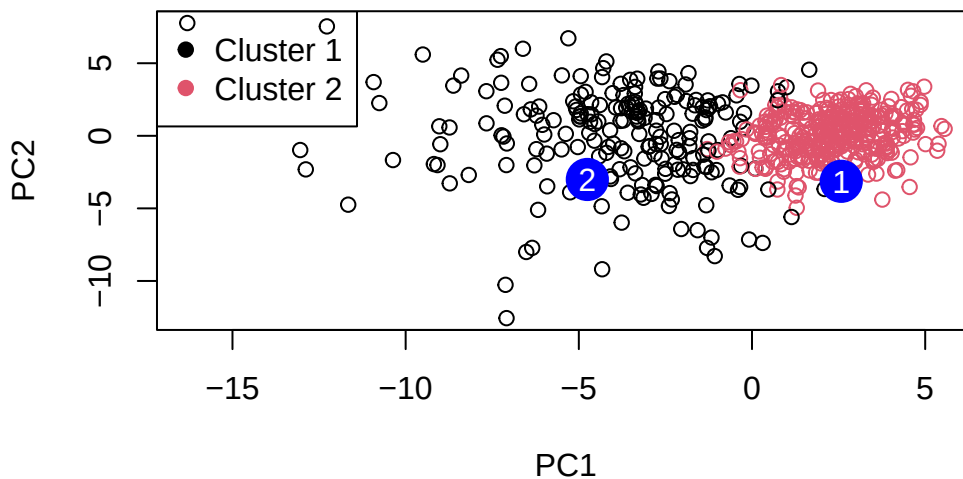
```

```

# Plot PC1 vs. PC2, colored by hclust clusters.
plot(wisc.pr$x[,1:2], col=wisc.pr.hclust.clusters)
legend("topleft", legend = c("Cluster 1", "Cluster 2"), col=1:2, pch=19)

# Add labeled points on above plot corresponding to observations in new dataset.
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")

```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize Patient 2 for follow-up. Patients 1 and 2 score very differently along the PC1 axis: patient 1 falls further right (positive PC1 score) and amongst members of

Cluster 2 from PCA-based hierarchical clustering (red points), whereas patient 2 falls further left (negative PC1 score) and amongst members of Cluster 1 (black points). As seen in the “Combining Methods” section, we saw that Cluster 1 largely corresponds to malignant samples and Cluster 2 largely corresponds to benign samples. Thus, the PCA model would predict that the Patient 2 sample is malignant and the Patient 1 sample is benign, meaning we should prioritize Patient 2 for follow-up.